

Eksamen

Kopiereg voorbehou

Examination

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Module EBN122
Elektrisiteit en Elektronika
11 November 2008

Module EBN122
Electricity and Electronics
11 November 2008



UNIVERSITEIT VAN PRETORIA
UNIVERSITY OF PRETORIA
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Eksamensinligting:

Exam information:

Maksimum punte: <i>Maximum marks:</i>	90	Volpunte: <i>Full marks:</i>	90
Duur van vraestel: <i>Duration of paper:</i>	180 minute <i>180 minutes</i>	Oopboek / toeboek: <i>Open / closed book:</i>	Toe <i>Closed</i>
Totale aantal bladsye (hierdie blad ingesluit): <i>Total number of pages (including this page):</i>	21		

BELANGRIK- IMPORTANT

- Die eksamenregulasies van die Universiteit van Pretoria geld.
The examination regulations of the University of Pretoria apply.
- Vrae van Afdeling A moet op die OPTIESE MERKBLAD (KANT 2) beantwoord word.
Questions from Section A must be answered on the OPTICAL MARKSHEET (SIDE 2).
- Vrae van Afdeling B se finale antwoorde moet op die ANTWOORDBLAD beantwoord word.
Final answers for questions from Section B must be answered on the ANSWER SHEET.
- Die eksamen-antwoordboek is slegs vir studente se eie gebruik, en word nie ingeneem of gemerk nie.
The exam script is only for the student's own use and will not be handed in or graded.

- Polêre vorm: $A \angle \theta$; Reghoekige vorm: $a + jb$
Polar form: $A \angle \theta$; Rectangular form: $a + jb$

Dosent(e): <i>Lecturer(s):</i>	1 J Joubert
	2 I Craig
	3 S Sinha
	4 E Kruger

Groephoof: Ek bevestig dat die vraestel die uitkomstes toets soos gespesifiseer in die studiehandleiding.
Group head: I confirm that the question paper evaluates the outcomes as specified in the study guide.

Groephoof: <i>Group Head:</i>	Prof J Joubert	Handtekening: <i>Signature:</i>	
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Afdeling A

Section A

Vrae van hierdie afdeling moet op die ANTWOORDBLAD geantwoord word. Kies die mees korrekte antwoord.

Questions from this section should be answered on the ANSWER SHEET. Choose the answer which is most correct.

VRAAG/QUESTION 1 [40]

Vraag 1.1. [1]

Die frekwensie van die seine wat geproduseer word in 'n elektriese stroombaan is in die omgewing van $f = 10^9$ Hz. Stroombaanteorie kan gebruik word vir bane

- A) kleiner as 30 cm
- B) groter as 5 cm
- C) kleiner as 3 cm
- D) groter as 10 cm
- E) Geen van die bogenoemdes nie

Question 1.1. [1]

The frequency of the signals produced in an electric circuit are in the region of $f = 10^9$ Hz. Circuit theory can be used for circuits

- A) smaller than 30 cm
- B) larger than 5 cm
- C) smaller than 3 cm
- D) larger than 10 cm
- E) None of the above

Vraag 1.2. [1]

Die stroom deur 'n element word getoon in Fig. Q1.2. Bepaal die totale lading wat deur die element gevloei het by $t = 3$ s. Aanvaar dat geen stroom gevloei het vir $t < 0$ nie.

- A) -20C
- B) 0C
- C) 10C
- D) 20C
- E) 45C
- F) 65C
- G) 75C

Question 1.2. [1]

The current through an element is shown in Fig. Q1.2. Determine the total charge that passed through the element at $t = 3$ s. Assume that no current passed through the element for $t < 0$.

- A) -20C
- B) 0C
- C) 10C
- D) 20C
- E) 45C
- F) 65C
- G) 75C

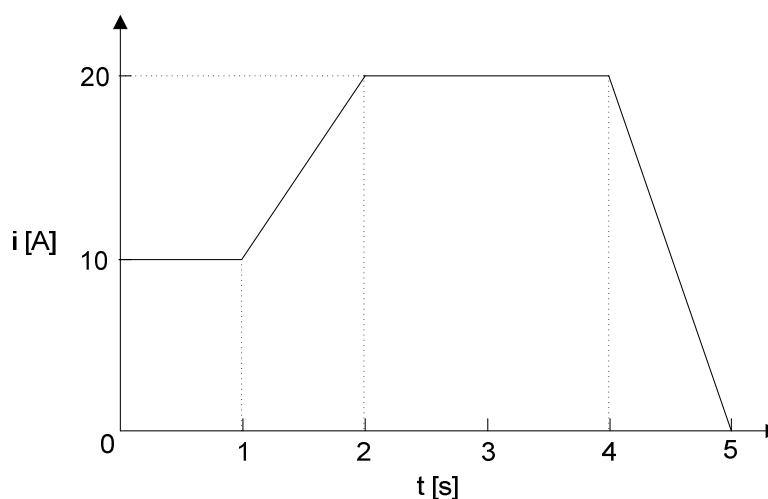


Fig. Q1.2

Vraag 1.3. [1]

Wanneer drywing in 'n element gemeet moet word deur van 'n voltmeter en 'n ammeter gebruik te maak, moet:

- A) Die voltmeter in serie en die ammeter in parallel verbind word.
- B) Die ammeter in serie en die voltmeter in parallel verbind word.
- C) Beide in serie verbind word.
- D) Beide in parallel verbind word.
- E) Geen van die bogenoemdes nie

Question 1.3. [1]

When measuring power in an element, by using a voltmeter and an ammeter:

- A) The voltmeter must be connected in serie and the ammeter in parallel.
- B) The ammeter must be connected in serie and the voltmeter in parallel.
- C) Both should be connected in serie.
- D) Both should be connected in parallel.
- E) None of the above.

Vraag 1.4. [1]

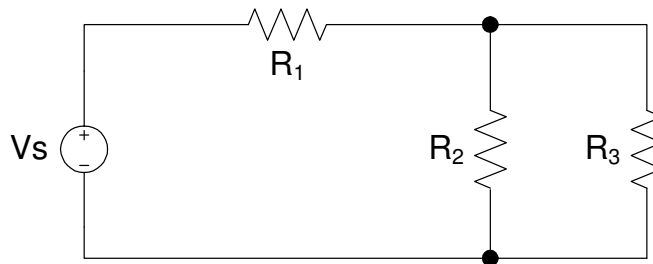
Vraag 1.4 verwys na Fig. Q1.4. In die stroombaan, sal 'n verlaging in R_3 lei tot 'n verlaging van:

- A) Die stroom deur R_3 .
- B) Die spanning oor R_1 .
- C) Die spanning oor R_3 .
- D) V_s .
- E) Geen van bogenoemde nie.

Question 1.4. [1]

Question 1.4 refers to Fig. Q1.4. In the circuit shown, a decrease in R_3 leads to a decrease of:

- A) The current through R_3 .
- B) The voltage across R_1 .
- C) The voltage across R_3 .
- D) V_s .
- E) None of the above.

**Fig. Q1.4****Vraag 1.5. [1]**

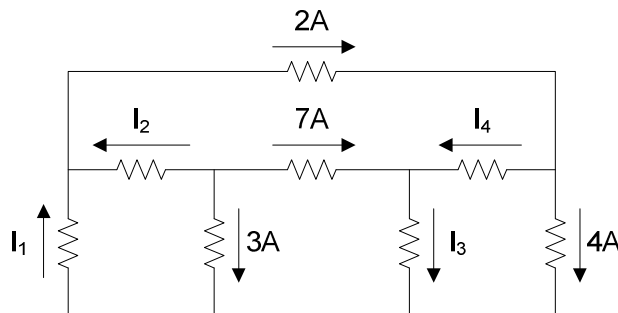
Vraag 1.5 verwys na Fig. Q1.5. Vir die stroombaan getoon:

- A) $I_1 = 8\text{A}$; $I_2 = 10\text{A}$
- B) $I_1 = 12\text{A}$; $I_2 = -10\text{A}$
- C) $I_1 = 12\text{A}$; $I_2 = 7\text{A}$
- D) $I_1 = 8\text{A}$; $I_2 = -7\text{A}$
- E) Geen van die bogenoemdes nie

Question 1.5. [1]

Question 1.5 refers to Fig. Q1.5. For the circuit shown:

- A) $I_1 = 8\text{A}$; $I_2 = 10\text{A}$
- B) $I_1 = 12\text{A}$; $I_2 = -10\text{A}$
- C) $I_1 = 12\text{A}$; $I_2 = 7\text{A}$
- D) $I_1 = 8\text{A}$; $I_2 = -7\text{A}$
- E) None of the above.

**Fig. Q1.5**

Vraag 1.6. [1]

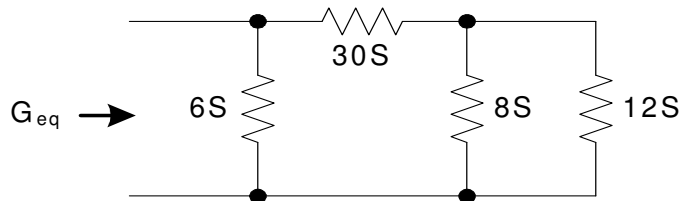
Vraag 1.6 verwys na Fig. Q1.6. Vir die stroombaan getoon, wat is die waarde van G_{eq} ?

- A) 2.788 S
- B) 6 S
- C) 10 S
- D) 18 S
- E) Geen van bogenoemde nie.

Question 1.6. [1]

Question 1.6 refers to Fig. Q1.6. For the circuit shown, what is the value of G_{eq} ?

- A) 2.788 S
- B) 6 S
- C) 10 S
- D) 18 S
- E) None of the above.

**Fig. Q1.6****Vraag 1.7. [1]**

Die omskakelingsreel vir Y na Δ netwerke is:

- A) Elke weerstand in die Δ network is die som van alle moontlike produkte van twee Y weerstande op 'n slag, gedeel deur die teenoorstaande Y weerstand.
- B) Elke weerstand in die Δ network is die produk van alle Y weerstande gedeel deur die teenoorstaande Y weerstand.
- C) Elke weerstand in die Δ network is die som van die Y weerstande gedeel deur die teenoorstaande Y weerstand.
- D) Elke weerstand in die Δ network is die produk van alle moontlike somme van twee Y weerstande op 'n slag, gedeel deur die teenoorstaande Y weerstand.
- E) Geen van die bogenoemdes nie

Question 1.7. [1]

The conversion rule for Y to Δ networks state that:

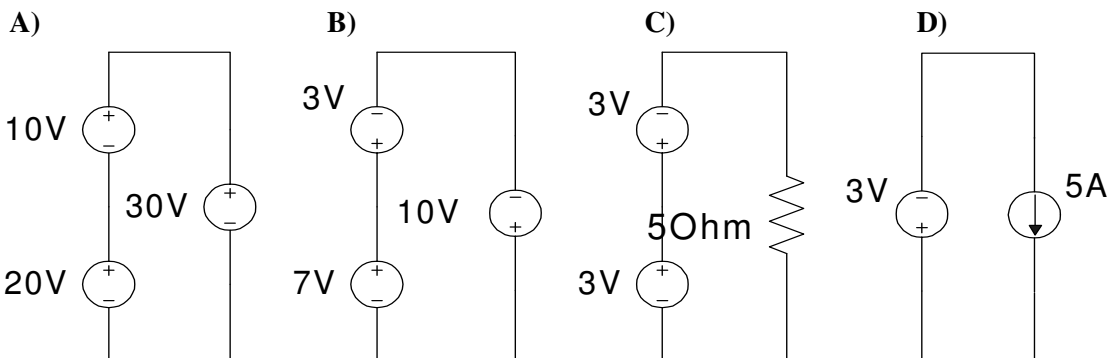
- A) Every resistor in the Δ network is the sum of all possible products of Y resistors taken two at a time, divided by the opposite Y resistor.
- B) Every resistor in the Δ network is the product of all Y resistors, divided by the opposite Y resistor.
- C) Every resistor in the Δ network is the sum of the Y resistors, divided by the opposite Y resistor.
- D) Every resistor in the Δ network is the product of all possible sums of Y resistors taken two at a time, divided by the opposite Y resistor.
- E) None of the above.

Vraag 1.8. [1]

Vraag 1.8 verwys na Fig. Q1.8. Watter van die volgende stroombane volg nie Kirchhoff se Spanningswet nie?

Question 1.8. [1]

Question 1.8 refers to Fig. Q1.8. Which of the following circuits does not follow Kirchhoff's Voltage Law?

**Fig. Q1.8**

Vraag 1.9. [1]

Wanneer 'n stroombron gedeel word deur twee kringe, vorm

- A) Een van die twee kringe 'n superkring.
- B) Beide die kringe superkringe.
- C) Die twee kringe en enige parallele elemente saam 'n superkring.
- D) Die twee kringe en enige serie elemente saam 'n superkring.
- E) Die twee kringe saam 'n superkring
- F) Geen van bogenoemde nie.

Question 1.9. [1]

When a current source is shared by two meshes, then

- A) One of the two meshes forms a supermesh.
- B) Both meshes form supermeshes.
- C) The two meshes and any parallel elements together form a supermesh.
- D) The two meshes and any series elements together form a supermesh.
- E) The two meshes together form a supermesh.
- F) None of the above.

Vraag 1.10. [1]

Vraag 1.10 verwys na Fig. Q1.10. Bepaal watter van die stroombane ekwivalent is.

- A) a en b
- B) b en d
- C) a en c
- D) c en d
- E) a en d
- F) Geen van die bogenoemdes nie

Question 1.10. [1]

Question 1.10 refers to Fig. Q1.10. Determine which pair of circuits are equivalent.

- A) a and b
- B) b and d
- C) a and c
- D) c and d
- E) a and d
- F) None of the above

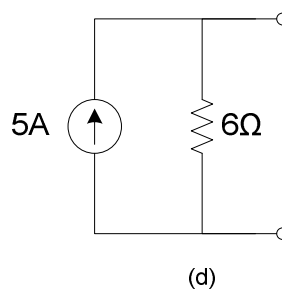
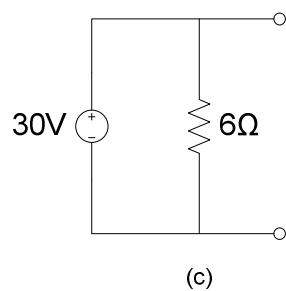
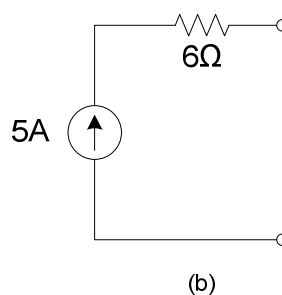
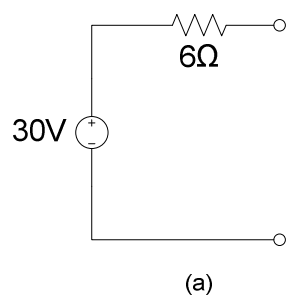


Fig. Q1.10

Vraag 1.11. [1]

Die spannings-stroom karakteristieke in Fig Q1.11b is verkry vanaf die onbekende lineêre stroombaan in Fig Q1.11a. Die baan bestaan slegs uit afhanklike en onafhanklike spanningsbronne, stroombronne en weerstande.

Bepaal die opebaan spanning V_{OC} ?

- A) 0 V
- B) 4 A
- C) 10 V
- D) 20 V
- E) 30 V
- F) 40 V
- G) Geen van die bogenoemdes nie

Question 1.11. [1]

The voltage-current characteristics in Fig Q1.11b were obtained from the unknown linear circuit in Fig Q1.11a. The circuit consists only of dependent and independent voltage sources, current sources and resistors.

What is the open circuit voltage V_{OC} ?

- A) 0 V
- B) 4 A
- C) 10 V
- D) 20 V
- E) 30 V
- F) 40 V
- G) None of the above

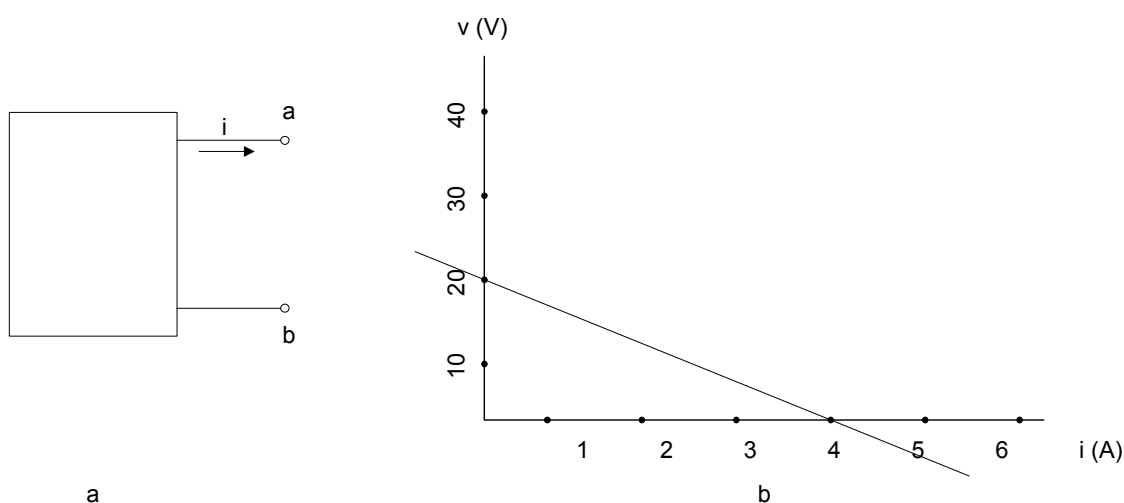


Fig. Q1.11

Vraag 1.12. [1]

Vraag 1.12 verwys na Fig. Q1.11. Bepaal die kort-gesluite stroom i_{SC} ?

- A) 0 A
- B) 1 A
- C) 2 A
- D) 3 A
- E) 4 A
- F) 5 A
- G) Geen van die bogenoemdes nie

Question 1.12. [1]

Question 1.12 refers to Fig. Q1.11. What is the short circuit current i_{SC} ?

- A) 0 A
- B) 1 A
- C) 2 A
- D) 3 A
- E) 4 A
- F) 5 A
- G) None of the above

Vraag 1.13. [1]

Vraag 1.13 verwys na Fig. Q1.11. Bepaal die Thévenin ekwivalente weerstand R_{TH} ?

- A) 6 Ω
- B) 5 Ω
- C) 4 Ω
- D) 3 Ω
- E) 2 Ω
- F) 1 Ω
- G) Geen van die bogenoemdes nie

Question 1.13. [1]

Question 1.13 refers to Fig. Q1.11. What is the Thévenin equivalent resistance R_{TH} ?

- A) 6 Ω
- B) 5 Ω
- C) 4 Ω
- D) 3 Ω
- E) 2 Ω
- F) 1 Ω
- G) None of the above

Vraag 1.14. [1]

Vraag 1.14 verwys na Fig. Q1.14. Bepaal die Norton ekwivalente weerstand van die stroombaan.

- A) $1\ \Omega$
- B) $2\ \Omega$
- C) $4\ \Omega$
- D) $6\ \Omega$
- E) $3\ \Omega$
- F) $5\ \Omega$
- G) Geen van die bogenoemdes nie

Question 1.14. [1]

Question 1.14 refers to Fig. Q1.14. Find the Norton equivalent resistance for the circuit.

- A) $1\ \Omega$
- B) $2\ \Omega$
- C) $4\ \Omega$
- D) $6\ \Omega$
- E) $3\ \Omega$
- F) $5\ \Omega$
- G) None of the above

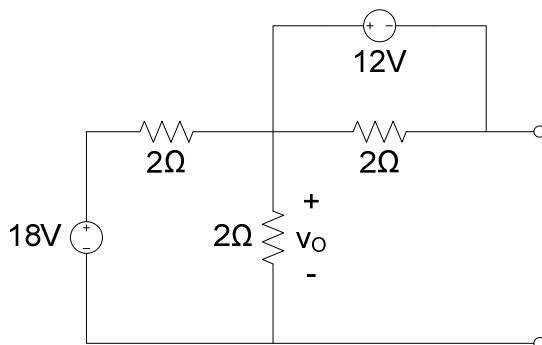


Fig. Q1.14

Vraag 1.15. [1]

Vraag 1.15 verwys na Fig. Q1.15. Bepaal die waarde van 'n resistor wat oor die oop terminale geplaas kan word wat maksimum drywingsoordrag sal verseker.

- A) $25\ \Omega$
- B) $12.5\ \Omega$
- C) $10\ \Omega$
- D) $8.11\ \Omega$
- E) $15\ \Omega$
- F) $9.21\ \Omega$
- G) Geen van die bogenoemdes nie

Question 1.15. [1]

Question 1.15 refers to Fig. Q1.15. Determine the value of a resistor placed across the open terminals resulting in maximum power transferred to the resistor.

- A) $25\ \Omega$
- B) $12.5\ \Omega$
- C) $10\ \Omega$
- D) $8.11\ \Omega$
- E) $15\ \Omega$
- F) $9.21\ \Omega$
- G) None of the above

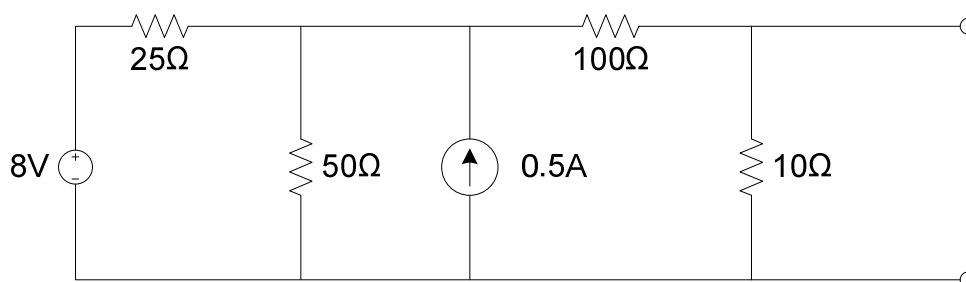


Fig. Q1.15

Vraag 1.16. [1]

Vraag 1.16 verwys na Fig. Q1.16. Vind die maksimum drywing wat deur die stroombaan gelewer kan word.

- A) 0.125 W
- B) 0.18 W
- C) 0.333 W
- D) 0.75 W
- E) 1.08 W
- F) Geen van die bogenoemdes nie

Question 1.16. [1]

Question 1.16 refers to Fig. Q1.16. Find the maximum power that can be delivered by the circuit.

- A) 0.125 W
- B) 0.18 W
- C) 0.333 W
- D) 0.75 W
- E) 1.08 W
- F) None of the above

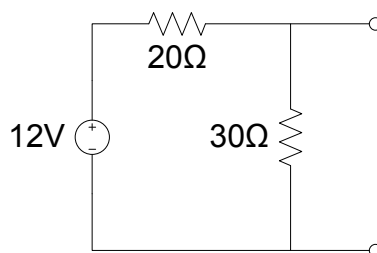


Fig. Q1.16

Vraag 1.17. [1]

Die ooplus-versterking van 'n ideale operasionele versterker is:

- A) 0
- B) Klein
- C) Baie groot
- D) Oneindig
- E) Afhanklik van die inset spannings
- F) Geen van die bogenoemdes nie

Question 1.17. [1]

The open-loop gain of an ideal operational amplifier is:

- A) 0
- B) Small
- C) Extremely large
- D) Infinite
- E) Dependant on input voltages
- F) None of the above

Vraag 1.18. [1]

Vraag 1.18 verwys na Fig. Q1.18. As $R_f = 10\text{ k}\Omega$, $R_1 = 40\text{ k}\Omega$, $R_2 = 500\text{ k}\Omega$, $R_3 = 250\text{ k}\Omega$ en $R_4 = 250\text{ k}\Omega$ is in die stroombaan, bepaal die analog uitset vir die binêre inset [0010].

- A) -0.35 V
- B) 0.35 V
- C) -0.04 V
- D) 0.04 V
- E) -0.31 V
- F) 0.31 V
- G) Geen van die bogenoemdes nie

Question 1.18. [1]

Question 1.18 refers to Fig. Q1.18. In the op-amp circuit, let $R_f = 10\text{ k}\Omega$, $R_1 = 40\text{ k}\Omega$, $R_2 = 500\text{ k}\Omega$, $R_3 = 250\text{ k}\Omega$ and $R_4 = 250\text{ k}\Omega$. Obtain the analogue output for a binary input of [0010].

- A) -0.35 V
- B) 0.35 V
- C) -0.04 V
- D) 0.04 V
- E) -0.31 V
- F) 0.31 V
- G) None of the above

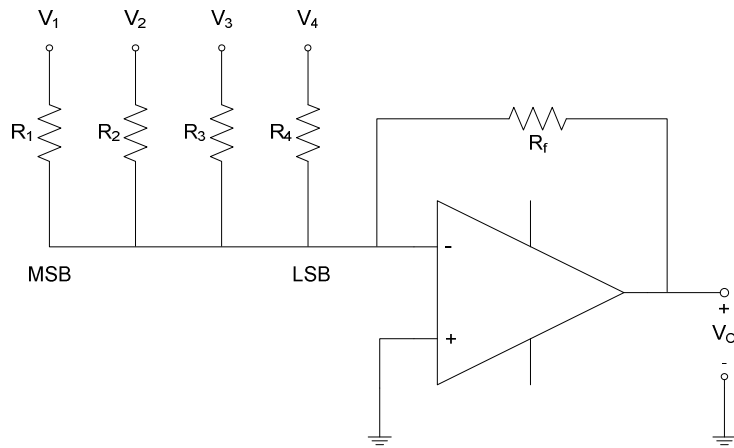


Fig. Q1.18

Vraag 1.19. [1]

Vraag 1.19 verwys na Fig. Q1.19. Bepaal die waarde van v_O .

- A) 0.8 V
- B) -3.2 V
- C) 6.4 V
- D) 3.2 V
- E) -6.4 V
- F) Geen van die bogenoemdes nie

Question 1.19. [1]

Question 1.19 refers to Fig. Q1.19. Calculate the value of v_O .

- A) 0.8 V
- B) -3.2 V
- C) 6.4 V
- D) 3.2 V
- E) -6.4 V
- F) None of the above

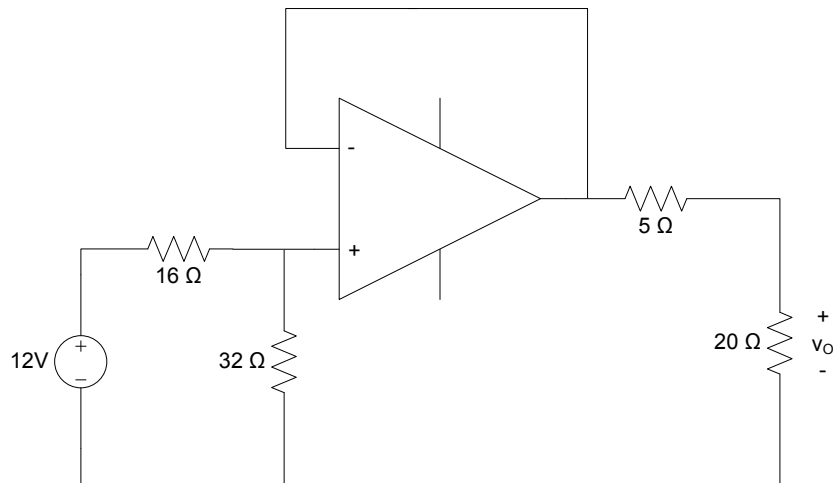


Fig. Q1.19

Vraag 1.20. [1]

Die intree-weerstande van 'n ideale operasionele versterker is:

- A) Albei nul.
- B) Albei oneindig.
- C) Die omkeer inset is nul en die nie-omkeer inset is oneindig.
- D) Die omkeer inset is oneindig en die nie-omkeer inset is nul.
- E) Die omkeer inset is baie groot en die nie-omkeer inset is nul.
- F) Geen van die bogenoemdes nie.

Question 1.20. [1]

The input resistances of an ideal operational amplifier are:

- A) Both zero.
- B) Both infinite.
- C) The inverting input is zero and the non-inverting input is infinite.
- D) The inverting input is infinite and the non-inverting input is zero.
- E) The inverting input is very large and the non-inverting input is zero.
- F) None of the above.

Vraag 1.21. [1]

Watter een van die volgende stellings in verband met kapasitors is vals?

- A) 'n Kapsitor bestaan uit twee geleidende plate wat geskei is deur 'n diëlektrikum.
- B) 'n Kapsitor is 'n oop baan vir DC.
- C) 'n Kapsitor kan gebruik word om fase te skuif.
- D) Die ekwivalente kapasitansie van parallel gekoppelde kapasitors is die som van die individuele kapasitansies.
- E) Geen van die bogenoemdes nie

Question 1.21. [1]

Which one of the statements regarding capacitors is false?

- A) A capacitor consists of two conducting plates separated by a dielectric.
- B) A capacitor is an open circuit to DC.
- C) A capacitor can be used to shift phase.
- D) The equivalent capacitance of parallel-connected capacitors is the sum of the individual capacitances.
- E) None of the above.

Vraag 1.22. [1]

Bereken die ekwivalente kapasitansie tussen terminale *a* en *b* in fig. Q1.22.

- A) $4.42 \mu\text{F}$
- B) $4.4 \mu\text{F}$
- C) $4.53 \mu\text{F}$
- D) $5 \mu\text{F}$
- E) $17.69 \mu\text{F}$
- F) $20 \mu\text{F}$
- G) Geen een van bogenoemdes

Question 1.22. [1]

Calculate the equivalent capacitance between terminals *a* and *b* in fig. Q1.22.

- A) $4.42 \mu\text{F}$
- B) $4.4 \mu\text{F}$
- C) $4.53 \mu\text{F}$
- D) $5 \mu\text{F}$
- E) $17.69 \mu\text{F}$
- F) $20 \mu\text{F}$
- G) None of the above

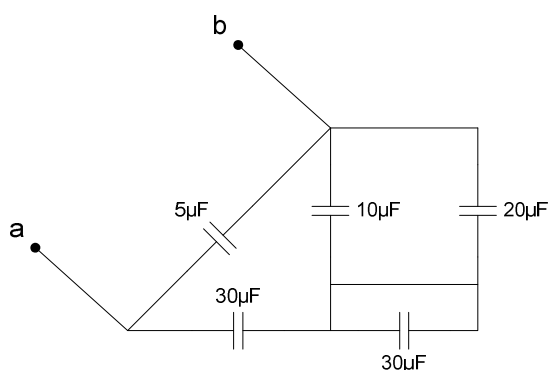


Fig. Q1.22

Vraag 1.23. [1]

Vraag 1.23 verwys na Fig. Q1.23. Bepaal die ekwivalente induktansie.

- A) 5.56 H
- B) 15 H
- C) 6.67 H
- D) 10 H
- E) 5 H
- F) Geen van die bogenoemdes nie

Question 1.23. [1]

Question 1.23 refers to Fig. Q1.23. Calculate the equivalent inductance.

- A) 5.56 H
- B) 15 H
- C) 6.67 H
- D) 10 H
- E) 5 H
- F) None of the above

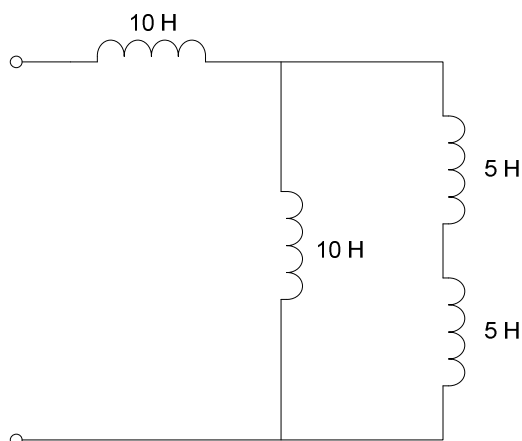


Fig. Q1.23

Vraag 1.24. [1]

Vraag 1.24 verwys na Fig. Q1.24. Bepaal v_L .

- A) 12 V
- B) 10 V
- C) 8 V
- D) 15 V
- E) 6.4 V
- F) Geen van die bogenoemdes nie

Question 1.24. [1]

Question 1.24 refers to Fig. Q1.24. Calculate the value of v_L .

- A) 12 V
- B) 10 V
- C) 8 V
- D) 15 V
- E) 6.4 V
- F) None of the above

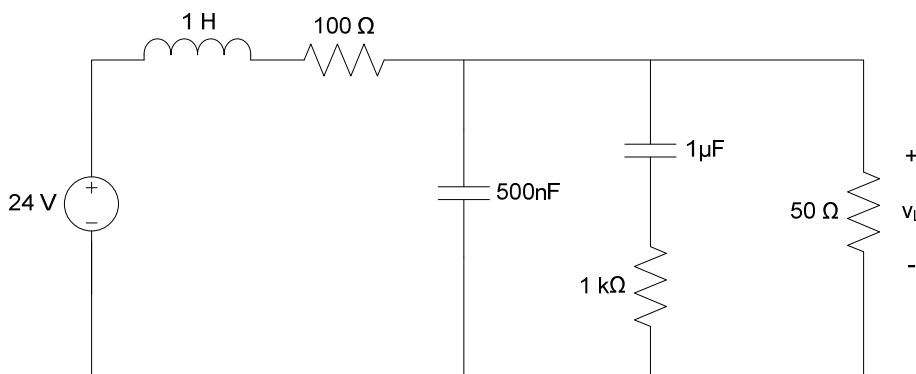


Fig. Q1.24

Vraag 1.25. [1] Watter een van die volgende stellings in verband met fasors is vals? A) As 'n sinusodale funksie gedifferensieer word is dit dieselfde as om sy fasor met $j\omega$ te vermenigvuldig. B) As 'n sinusodale funksie geïntegreer word is dit dieselfde as om sy fasor met $j\omega$ te deel. C) As twee sinusodale funksies opgetel word is dit dieselfde as om hulle fasors op te tel. D) 'n Fasor is 'n komplekse getal wat die amplitude en frekwensie van 'n sinusodale funksie aandui. E) Geen van die bogenoemdes nie.	Question 1.25. [1] Which one of the statements regarding phasors is false? A) Differentiating a sinusoid is equivalent to multiplying its corresponding phasors by $j\omega$ B) Integrating a sinusoid is equivalent to dividing its corresponding phasors by $j\omega$ C) Adding sinusoids of the same frequency is equivalent to adding their corresponding phasors. D) A phasor is a complex number that represents the amplitude and frequency of a sinusoid E) None of the above
Vraag 1.26. [1] Skryf die volgende sinusodale funksies as fasors: (a) $v = -10\cos(2t + 80^\circ)$ V (b) $i = 15\sin(10t + 25^\circ)$ A A) $V = 10\angle -100^\circ$, $I = 15\angle -65^\circ$ B) $V = 10\angle 260^\circ$, $I = 15\angle 115^\circ$ C) $V = 10\angle 260^\circ$, $I = 15\angle -65^\circ$ D) $V = 10\angle -100^\circ$, $I = 15\angle 115^\circ$ E) $V = -10\angle 80^\circ$, $I = 15\angle 25^\circ$ F) $V = -10\angle -100^\circ$, $I = -15\angle -65^\circ$ G) Geen van die bogenoemdes nie	Question 1.26. [1] Express these sinusoids as phasors: (a) $v = -10\cos(2t + 80^\circ)$ V (b) $i = 15\sin(10t + 25^\circ)$ A A) $V = 10\angle -100^\circ$, $I = 15\angle -65^\circ$ B) $V = 10\angle 260^\circ$, $I = 15\angle 115^\circ$ C) $V = 10\angle 260^\circ$, $I = 15\angle -65^\circ$ D) $V = 10\angle -100^\circ$, $I = 15\angle 115^\circ$ E) $V = -10\angle 80^\circ$, $I = 15\angle 25^\circ$ F) $V = -10\angle -100^\circ$, $I = -15\angle -65^\circ$ G) None of the above.
Vraag 1.27. [1] Evalueer die volgende komplekse getal en skryf die antwoord in reghoekige vorm. $3 + (5\angle 30^\circ)(9 - j5)$ A) $54.47 + j0.849$ B) $47.15 - j26.46$ C) $47.15 + j26.46$ D) $-44.15 - j26.46$ E) Geen van die bogenoemdes nie	Question 1.27. [1] Evaluate the following complex number and express in rectangular form. $3 + (5\angle 30^\circ)(9 - j5)$ A) $54.47 + j0.849$ B) $47.15 - j26.46$ C) $47.15 + j26.46$ D) $-44.15 - j26.46$ E) None of the above

Vraag 1.28. [1]

Die impedansie en admitansie van 'n 1mF kapisitor, wat teen $\omega = 100$ rad/s werk, is:

- A) $0.1 \angle -90^\circ \Omega$ and $10 \angle 90^\circ S$
- B) $10 \angle -90^\circ \Omega$ and $0.1 \angle 90^\circ S$
- C) $0.1 \angle -90^\circ S$ and $10 \angle 90^\circ \Omega$
- D) $10 \angle 90^\circ \Omega$ and $0.1 \angle -90^\circ S$
- E) Geen van die bogenoemdes nie

Question 1.28. [1]

The impedance and admittance of a 1mF capacitor operating at $\omega = 100$ rad/s are:

- A) $0.1 \angle -90^\circ \Omega$ and $10 \angle 90^\circ S$
- B) $10 \angle -90^\circ \Omega$ and $0.1 \angle 90^\circ S$
- C) $0.1 \angle -90^\circ S$ and $10 \angle 90^\circ \Omega$
- D) $10 \angle 90^\circ \Omega$ and $0.1 \angle -90^\circ S$
- E) None of the above

Vraag 1.29. [1]

Die impedansie en admitansie van 'n 10mH inductor, wat teen $\omega = 1000$ rad/s werk, is:

- A) $0.1 \angle -90^\circ \Omega$ and $10 \angle 90^\circ S$
- B) $10 \angle -90^\circ \Omega$ and $0.1 \angle 90^\circ S$
- C) $0.1 \angle -90^\circ S$ and $10 \angle 90^\circ \Omega$
- D) $10 \angle 90^\circ \Omega$ and $0.1 \angle -90^\circ S$
- E) Geen van die bogenoemdes nie

Question 1.29. [1]

The impedance and admittance of a 10mH inductor operating at $\omega = 1000$ rad/s are:

- A) $0.1 \angle -90^\circ \Omega$ and $10 \angle 90^\circ S$
- B) $10 \angle -90^\circ \Omega$ and $0.1 \angle 90^\circ S$
- C) $0.1 \angle -90^\circ S$ and $10 \angle 90^\circ \Omega$
- D) $10 \angle 90^\circ \Omega$ and $0.1 \angle -90^\circ S$
- E) None of the above

Vraag 1.30. [1]

Vraag 1.30 verwys na Fig. Q1.30. Bepaal $i(t)$.

- A) $2\sin(6280t-89^\circ)A$
- B) $2\sin(6280t+1^\circ)A$
- C) $1\cos(6280t-89^\circ)A$
- D) $1\sin(6280t+1^\circ)A$
- E) Geen van die bogenoemdes nie.

Question 1.30. [1]

Question 1.30 refers to Fig. Q1.30. Find $i(t)$.

- A) $2\sin(6280t-89^\circ)A$
- B) $2\sin(6280t+1^\circ)A$
- C) $1\cos(6280t-89^\circ)A$
- D) $1\sin(6280t+1^\circ)A$
- E) None of the above

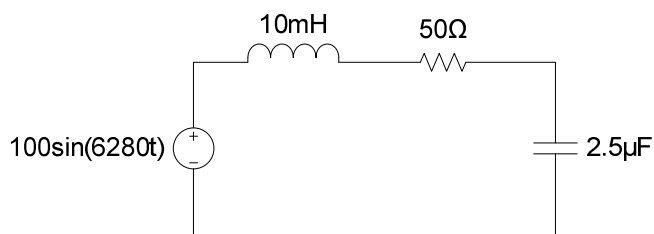


Fig. Q1.30

Vraag 1.31. [2]

Die stroom deur 'n 10mH inductor is $i(t) = 34.5\cos(314.2t+12^\circ)A$. As die spanning oor die inductor as $V_m\cos(\omega t+\Phi)V$ gegee word, vind V_m , ω , and Φ .

- A) 108.4V, 314.2rad/s, 102°
- B) 96.5V, 314.2rad/s, 102°
- C) 96.5V, 314.2rad/s, -78°
- D) 108.4V, 50rad/s, -78°
- E) Geen van die bogenoemdes nie.

Question 1.31. [2]

Let the current into a 10mH inductor be given by $i(t) = 34.5\cos(314.2t+12^\circ)A$. If the voltage across the inductor is given by $V_m\cos(\omega t+\Phi)V$, find V_m , ω , and Φ .

- A) 108.4V, 314.2rad/s, 102°
- B) 96.5V, 314.2rad/s, 102°
- C) 96.5V, 314.2rad/s, -78°
- D) 108.4V, 50rad/s, -78°
- E) None of the above

Vraag 1.32. [2]

Vraag 1.32 verwys na Fig. Q1.32. Bepaal die waarde van die stroom I_0 .

- A) $4\angle 0^\circ$ A
- B) $2.4\angle -90^\circ$ A
- C) $0.6\angle 0^\circ$ A
- D) -1 A
- E) Geen van die bogenoemdes nie

Question 1.32. [2]

Question 1.32 refers to Fig. Q1.32. The value of the current I_0 is:

- A) $4\angle 0^\circ$ A
- B) $2.4\angle -90^\circ$ A
- C) $0.6\angle 0^\circ$ A
- D) -1 A
- E) None of the above.

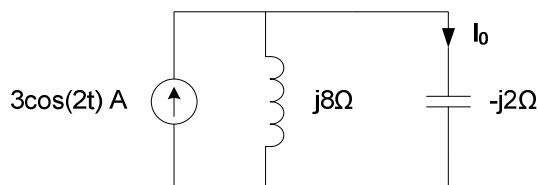


Fig. Q1.32

Vraag 1.33. [2]

Vraag 1.33 verwys na Fig. Q1.33. Bereken V_0 deur van nodale analise gebruik te maak.

- A) $2.22\angle 0^\circ$ A
- B) $2.97\angle -42^\circ$ A
- C) $3.98\angle -5.71^\circ$ A
- D) $3.98\angle 5.71^\circ$ A
- E) Geen van die bogenoemdes nie

Question 1.33. [2]

Question 1.33 refers to Fig. Q1.33. Solve V_0 using nodal analysis.

- A) $2.22\angle 0^\circ$ A
- B) $2.97\angle -42^\circ$ A
- C) $3.98\angle -5.71^\circ$ A
- D) $3.98\angle 5.71^\circ$ A
- E) None of the above.

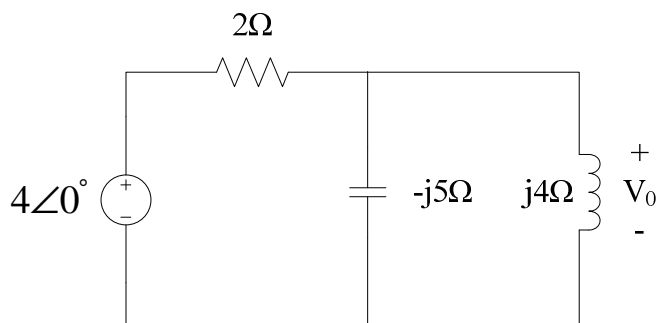


Fig. Q1.33

Vraag 1.34. [2]

Vraag 1.34 verwys na Fig. Q1.34. Bepaal die opebaan spanning $V_{OB} = V_{ab}$.

- A) $80 \angle 0^\circ \text{ V}$
- B) $20 \angle 0^\circ \text{ V}$
- C) $1.94 \angle -14.04^\circ \text{ V}$
- D) 3 V
- E) Geen van die bogenoemdes nie

Question 1.34. [2]

Question 1.34 refers to Fig. Q1.34. Determine the open-circuit voltage $V_{OC} = V_{ab}$.

- A) $80 \angle 0^\circ \text{ V}$
- B) $20 \angle 0^\circ \text{ V}$
- C) $1.94 \angle -14.04^\circ \text{ V}$
- D) 3 V
- E) None of the above

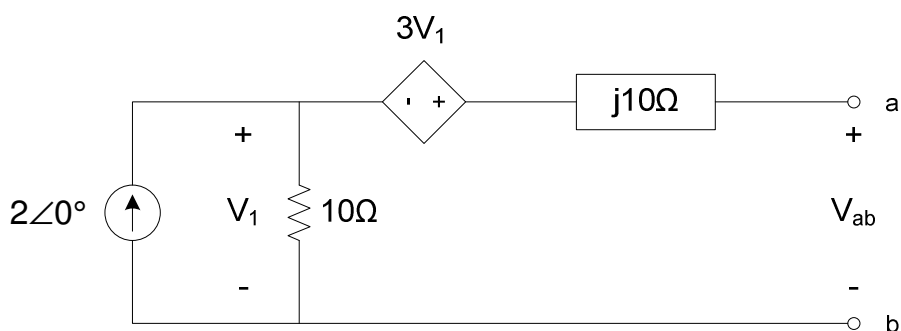


Fig. Q1.34

Vraag 1.35. [2]

Vraag 1.35 verwys na Fig. Q1.34. Bepaal die geslotebaan stroom $I_{GB} = I_{ab}$.

- A) $2 \angle 0^\circ \text{ A}$
- B) $0.2 \angle 0^\circ \text{ A}$
- C) $1.94 \angle -14.04^\circ \text{ A}$
- D) $14.04 \angle 1.94^\circ \text{ A}$
- E) Geen van die bogenoemdes nie

Question 1.35. [2]

Question 1.35 refers to Fig. Q1.34. Determine the short-circuit current $I_{SC} = I_{ab}$.

- A) $2 \angle 0^\circ \text{ A}$
- B) $0.2 \angle 0^\circ \text{ A}$
- C) $1.94 \angle -14.04^\circ \text{ A}$
- D) $14.04 \angle 1.94^\circ \text{ A}$
- E) None of the above

Afdeling B

Section B

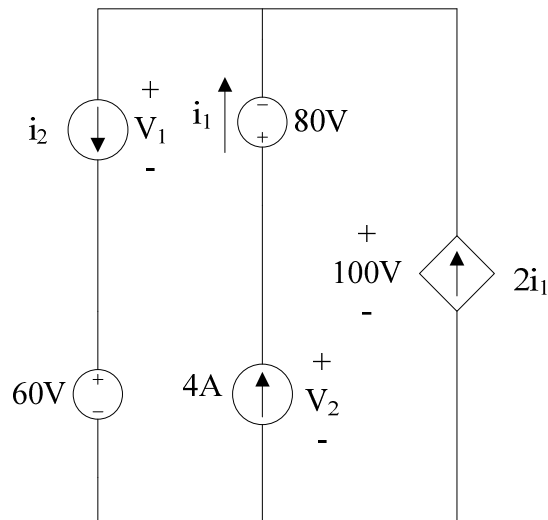
Die eksamenboek is slegs vir die student se eie gebruik - finale antwoorde van vrae van hierdie afdeling moet na die ANTWOORDBLAD oorgedra word.

The examination script is only for the student's own use - final answers of questions from this section should be transferred to the ANSWER SHEET.

VRAAG 1 – QUESTION 1 [12]

Vraag 1.1

Question 1.1



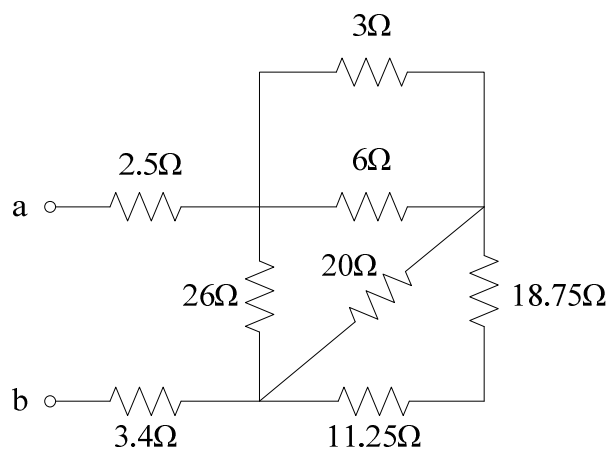
- a) Bepaal die stroom i_1 .
b) Bepaal die totale drywing deur die stroombaan gelewer.

- a) Determine the current i_1 .
b) Find the total power delivered in the circuit.

[3]

Vraag 1.2

Question 1.2



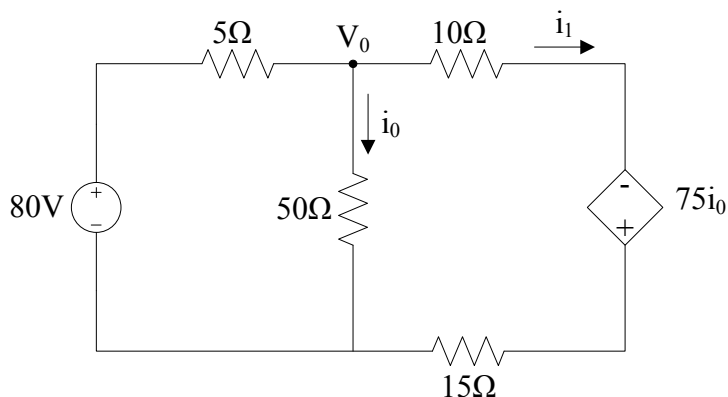
Bereken die ekwivalente weerstand R_{ab} .

Find the equivalent resistance R_{ab} .

[2]

Vraag 1.3

Question 1.3



Gebruik nodale analyse om die stroombaan op te los.

- Bereken die waarde van v_0 .
- Bereken die drywing wat deur die afhanklike spanningsbron gelewer word.

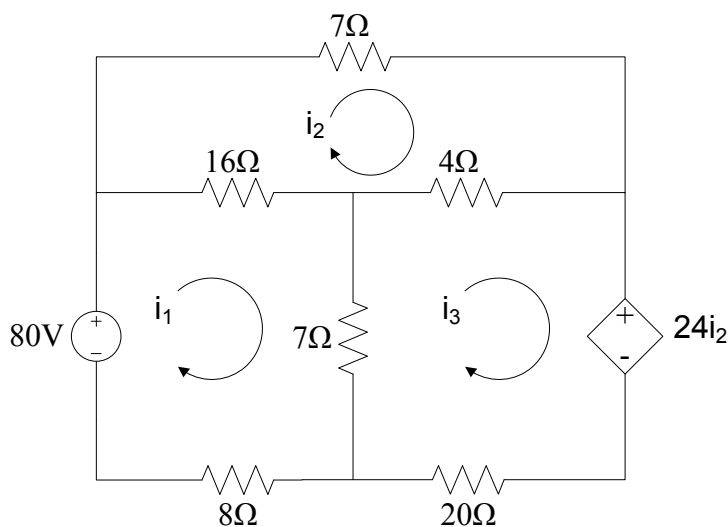
Use the node-voltage method to solve the circuit.

- Determine the value of v_0
- Determine the power delivered by the dependent voltage source.

[3]

Vraag 1.4

Question 1.4



Gebruik lus-analise om 'n matriksvergelyking van die volgende vorm op te stel:

$$\begin{bmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & 27 & a_{23} \\ a_{31} & a_{32} & 31 \end{bmatrix} \begin{bmatrix} i_1 \\ i_2 \\ i_3 \end{bmatrix} = \begin{bmatrix} 80 \\ b_2 \\ b_3 \end{bmatrix}$$

Los die vergelyking op om i_1 te bepaal.

Use the mesh-current method to set up a matrix equation of the following form:

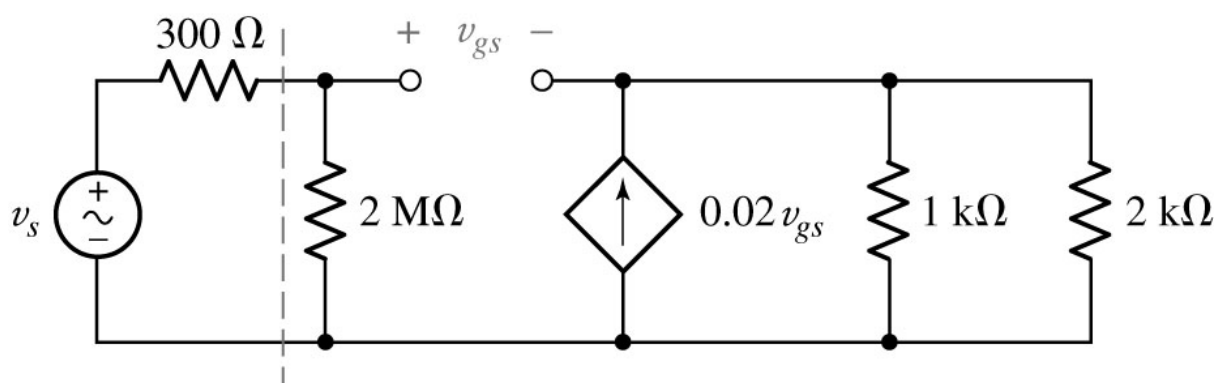
$$\begin{bmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & 27 & a_{23} \\ a_{31} & a_{32} & 31 \end{bmatrix} \begin{bmatrix} i_1 \\ i_2 \\ i_3 \end{bmatrix} = \begin{bmatrix} 80 \\ b_2 \\ b_3 \end{bmatrix}$$

Solve the equation to determine i_1 .

[4]

VRAAG 2 – QUESTION 2 [15]

Vraag 2.1	Question 2.1
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Bepaal die Thévenin ekwivalente weerstand vir die stroombaan regs van die stippellyn.

Determine the Thévenin equivalent resistance of the circuit to the right of the dashed line.

[2]

Vraag 2.2	Question 2.2
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'n Sekere praktiese gelykspanningsbron kan 'n stroom van 2.5A lewer as dit (oombliklik) gekortsluit word, en kan ook 'n drywing van 80W aan 'n 20Ω las lewer. Bepaal:

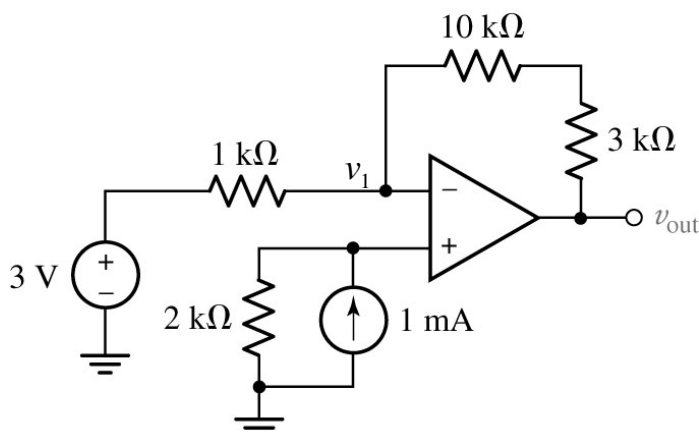
- (a) opebaan spanning,
- (b) maksimum drywing wat aan 'n goed geselekteerde lasweerstand, R_L gelever kan word, en
- (c) die waarde van R_L (wat maksimum drywingsoordrag verseker).

A certain practical DC voltage source can provide a current of 2.5A when it is (momentarily) short-circuited, and can also provide a power of 80W to a 20Ω load. Determine:

- (a) open circuit voltage,
- (b) maximum power it could deliver to a well-chosen load resistor, R_L , and
- (c) the value of R_L (enabling maximum power transfer to the load).

[4]

Vraag 2.3	Question 2.3
-----------	--------------



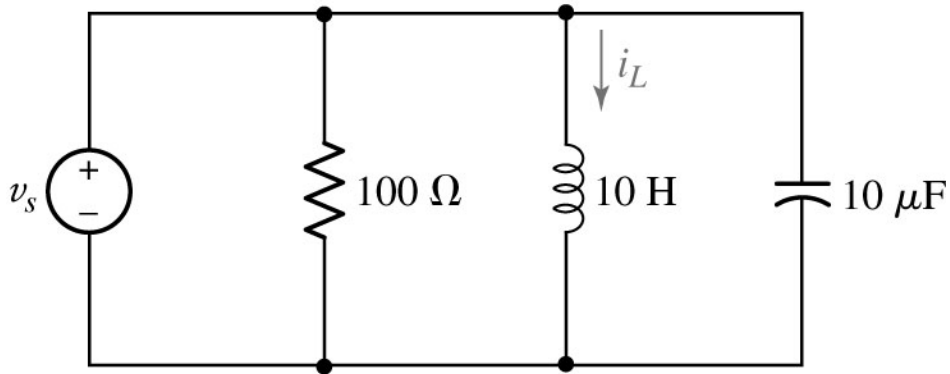
Aanvaar die op-amp is ideaal. Bereken v_1 en v_{out} .

Assuming the op amp is ideal, determine v_1 and v_{out} .

[3]

Vraag 2.4

Question 2.4



Laat $v_s = 400t^2$ V for $t > 0$ en $i_L(0) = 0.5$ A. Teen $t = 0.4$ s, bepaal die volgende energiewaardes:

- (a) gestoor in die kapasitor,
- (b) gestoor in die inductor, en
- (c) verkwis in die weerstand vanaf $t = 0$.

Let $v_s = 400t^2$ V for $t > 0$ and $i_L(0) = 0.5$ A. At $t = 0.4$ s, determine the following energy values:

- (a) stored in the capacitor,
- (b) stored in the inductor, and
- (c) dissipated by the resistor since $t = 0$.

[6]

VRAAG 3 – QUESTION 3 [23]

Vraag 3.1

Question 3.1

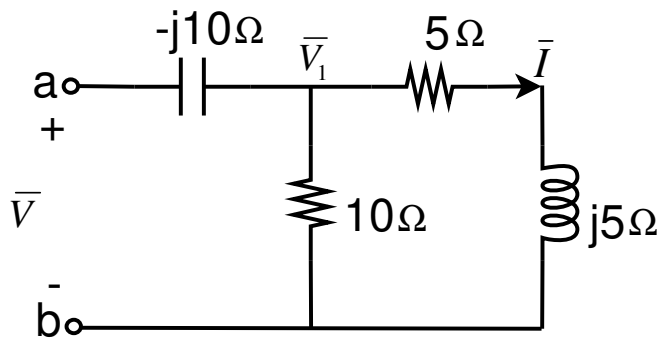
Teen $\omega = 2$ rad/s het 'n 3Ω weerstand in serie met 'n 2 H induktor presies dieselfde impedansie as 'n weerstand R in parallel met 'n induktor L . Bepaal R en L .

At $\omega = 2$ rad/s a 3Ω resistor in series with a 2 H inductor has exactly the same impedance as a resistor R in parallel with an inductor L . Determine R en L .

[2]

Vraag 3.2

Question 3.2



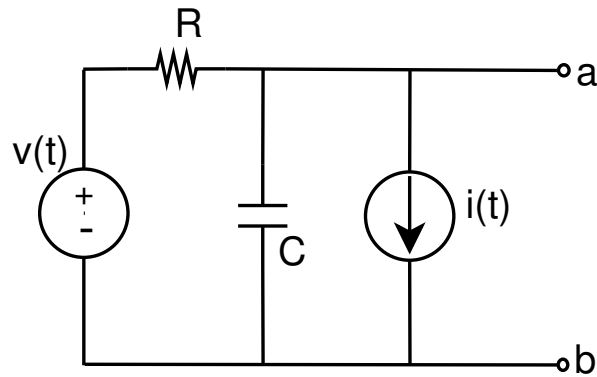
- (a) Bepaal die intree-impedansie by terminale a-b. (reghoekige vorm)
- (b) Indien 'n spanning $\bar{V} = 100\angle 0^\circ$ V aangelê word by terminale a-b, bepaal $\bar{V}_1$. (polêre vorm)
- (c) Indien 'n stroombron $\bar{I}_s = 2\angle -90^\circ$ A aangelê word by terminale a-b (die stroom vloei in by terminaal a), bepaal $\bar{I}$. (polêre vorm)

- (a) Determine the input impedance at terminals a-b. (rectangular form)
- (b) If a voltage $\bar{V} = 100\angle 0^\circ$ V is applied at terminals a-b, determine $\bar{V}_1$. (polar form)
- (c) If a current source $\bar{I}_s = 2\angle -90^\circ$ A is applied at terminals a-b (the current enters terminal a), determine $\bar{I}$. (polar form)

[6]

Vraag 3.3

Question 3.3



Bepaal die Thevenin ekwivalente baan (in die fasor-domein) mbt terminale a-b indien $v(t) = 5\cos(40t)$ V, $i(t) = 0.2\cos(40t)$ A, $R = 10\ \Omega$ en $C = 1/400$ F.

(Bereken $\bar{V}_{th}$ en $\bar{Z}_{th}$ in polêre vorm).

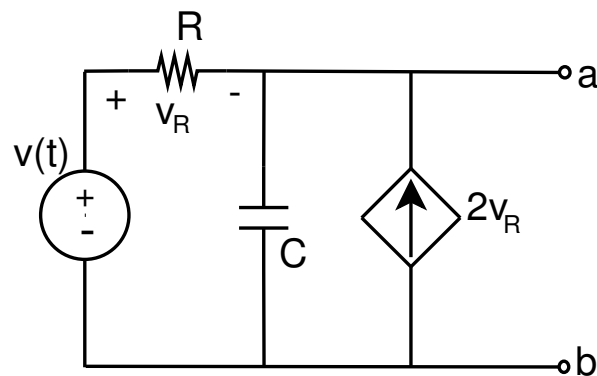
Determine the Thevenin equivalent circuit (in the phasor domain) wrt terminals a-b if $v(t) = 5\cos(40t)$ V, $i(t) = 0.2\cos(40t)$ A, $R = 10\ \Omega$ and $C = 1/400$ F.

(Compute $\bar{V}_{th}$ and $\bar{Z}_{th}$ in polar form).

[4]

Vraag 3.4

Question 3.4



Bepaal die Norton ekwivalente baan mbt terminale a-b indien $v(t) = 5\cos(40t)$ V, $R = 10\ \Omega$ en $C = 1/400$ F.

(Bereken $\bar{I}_N$ en $\bar{Z}_N$ in polêre vorm).

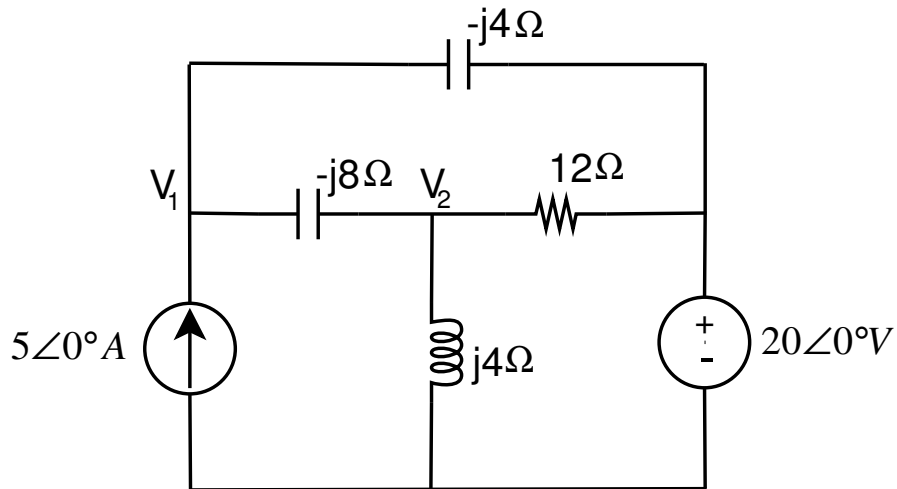
Determine the Norton equivalent circuit wrt terminals a-b if $v(t) = 5\cos(40t)$ V, $R = 10\ \Omega$ and $C = 1/400$ F.

(Compute $\bar{I}_N$ and $\bar{Z}_N$ in polar form).

[4]

Vraag 3.5

Question 3.5



Gebruik node-analise om twee gelyktydige vergelykings vir fasors $\bar{V}_1$ en $\bar{V}_2$ op te stel. Skryf in die vorm

$$\bar{a}\bar{V}_1 + \bar{b}\bar{V}_2 = 40 - j40$$

$$\bar{c}\bar{V}_1 + \bar{d}\bar{V}_2 = 40$$

met $\bar{a}$, $\bar{b}$, $\bar{c}$ en $\bar{d}$ komplekse koeffisiënte in reghoekige vorm.

Use nodal analysis to set up two simultaneous equations for phasors $\bar{V}_1$ and $\bar{V}_2$. Write in the form

$$\bar{a}\bar{V}_1 + \bar{b}\bar{V}_2 = 40 - j40$$

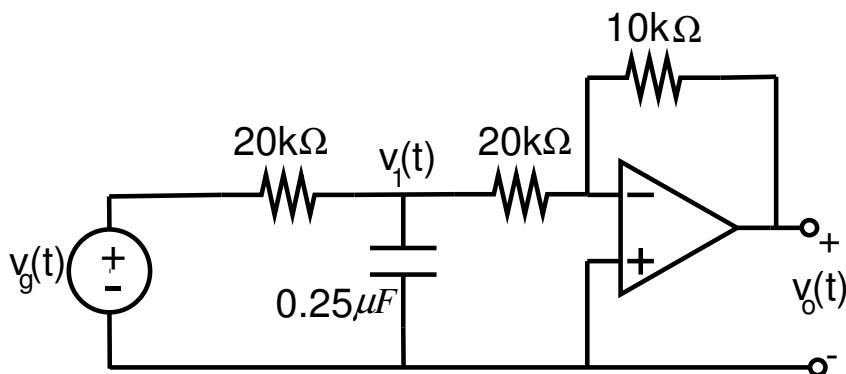
$$\bar{c}\bar{V}_1 + \bar{d}\bar{V}_2 = 40$$

with $\bar{a}$, $\bar{b}$, $\bar{c}$ and $\bar{d}$ complex coefficients in rectangular form.

[4]

Vraag 3.6

Question 3.6



Indien $v_g(t) = 4\cos(200t)$ V, bepaal $v_1(t)$ en $v_0(t)$.

If $v_g(t) = 4\cos(200t)$ V, determine $v_1(t)$ and $v_0(t)$.

[3]

EINDE - END

EBN122 EKSAMEN / EXAMINATION 2008
ANTWOORDBLAD / ANSWER SHEET

Student se besonderhede/Student's details		
Van en voorletters: <i>Surname and initials:</i>		Studentenommer: <i>Student number:</i>
Graadkursus: <i>Degree Course:</i>		
Tel. nr.: <i>Tel no.:</i>		

Afdeling A Section A	Vraag 1/ Question 1	/40
Afdeling B Section B	Vraag 1/ Question 1	/12
	Vraag 2/ Question 2	/15
	Vraag 3/ Question 3	/23
Totaal/Total:		/90

AFDELING B / SECTION B

VRAAG 1 – QUESTION 1		
1.1	$i_1 =$ [1]	$P_{\text{total}} =$ [2]
1.2	$R_{\text{ab}} =$ [2]	
1.3	$v_0 =$ [1]	$P_{\text{del}} =$ [2]
1.4	$\begin{bmatrix} & & \\ & 27 & \\ & & 31 \end{bmatrix} \begin{bmatrix} i_1 \\ i_2 \\ i_3 \end{bmatrix} = \begin{bmatrix} 80 \\ \\ \end{bmatrix}$	
	$i_1 =$ [1]	

VRAAG 2 – QUESTION 2			
2.1	$R_{th} =$ [2]		
2.2	$V_{oc} =$ [2]	$P_{max} =$ [1]	$R_L =$ [1]
2.3	$v_1 =$ [1]	$v_{out} =$ [2]	
2.4	$w_C =$ [2]	$w_L =$ [2]	$w_R =$ [2]

VRAAG 3 – QUESTION 3			
3.1	$R =$ [1]	$L =$ [1]	
3.2	$\bar{Z}_{in} =$ [2]		
	$\bar{V}_1 =$ [2]		
	$\bar{I} =$ [2]		
3.3	$\bar{V}_{th} =$ [2]	$\bar{Z}_{th} =$ [2]	
3.4	$\bar{I}_N =$ [2]	$\bar{Z}_N =$ [2]	
3.5	$\bar{a} =$ [1]	$\bar{b} =$ [1]	
	$\bar{c} =$ [1]	$\bar{d} =$ [1]	
3.6	$v_1(t) =$ [2]	$v_0(t) =$ [1]	